

3 System and memory overview

3.1 System architecture

The STM32F302xB/C/D/E, STM32F358xC and STM32F398xE main system consists of:

- Five masters:
 - Cortex[®]-M4 core I-bus
 - Cortex[®]-M4 core D-bus
 - Cortex[®]-M4 core S-bus
 - GP-DMA1 and GP-DMA2 (general-purpose DMAs)
- Seven (eight in STM32F303xDxE and STM32F398xE) slaves:
 - Internal Flash memory on the DCode
 - Internal flash memory on ICode
 - Up to internal 40-Kbyte SRAM
 - Internal 8-Kbyte CCM SRAM (16-Kbyte CCM SRAM for STM32F303xE and STM32F398xE)
 - FMC in STM32F303xDxE and STM32F398xE
 - AHB to APBx (APB1 or APB2), which connect all the APB peripherals
 - AHB dedicated to GPIO ports
 - ADCs 1, 2, 3 and 4.

The STM32F303x6/8 and STM32F328x8 main system consists of:

- Four masters:
 - Cortex[®]-M4 core I-bus
 - Cortex[®]-M4 core D-bus
 - Cortex[®]-M4 core S-bus
 - DMA1 (general-purpose DMA)
- Seven slaves:
 - Internal flash memory on the DCode
 - Internal flash memory on ICode
 - Up to 12-Kbyte internal SRAM
 - Internal 4-Kbyte CCM SRAM
 - AHB to APBx (APB1 or APB2), which connect all the APB peripherals
 - AHB dedicated to GPIO ports
 - ADCs 1 and 2

The interconnection uses a multilayer AHB bus architecture as shown in figures [1](#) to [3](#).

Figure 1. STM32F303xB/C and STM32F358xC system architecture

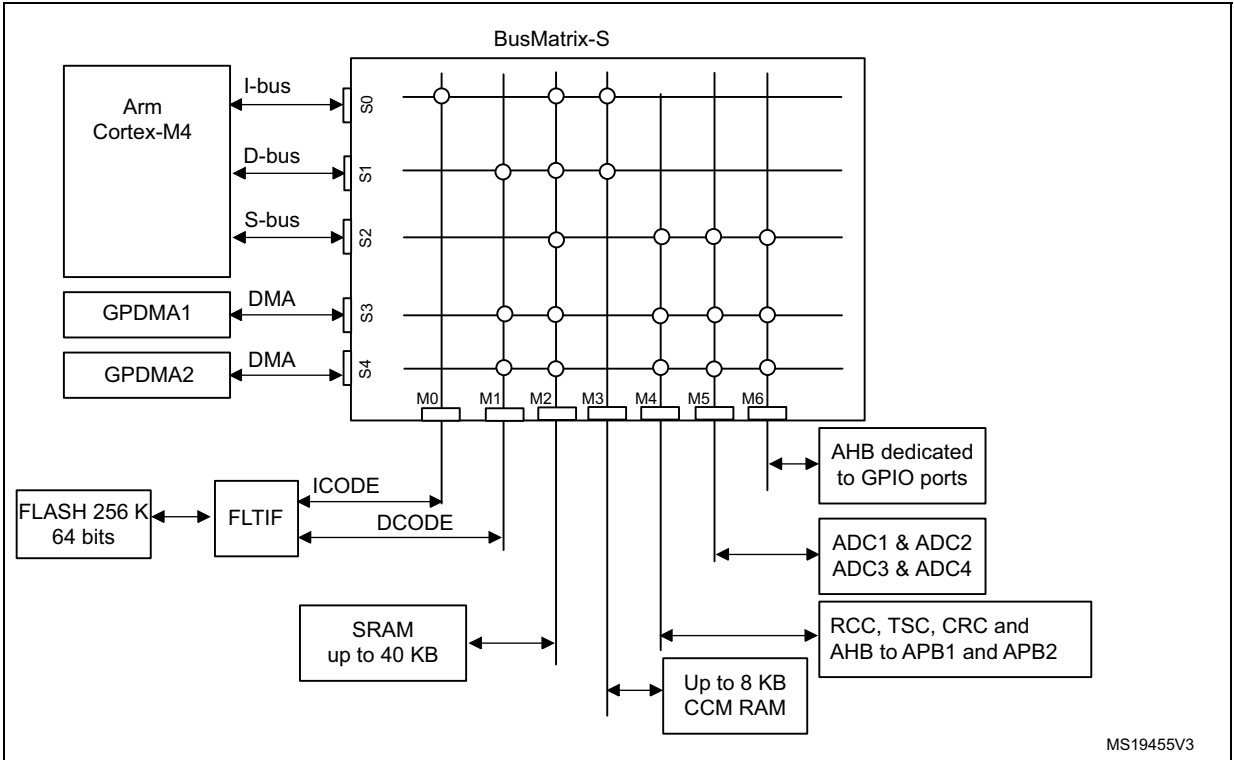


Figure 2. STM32F303x6/8 and STM32F328x8 system architecture

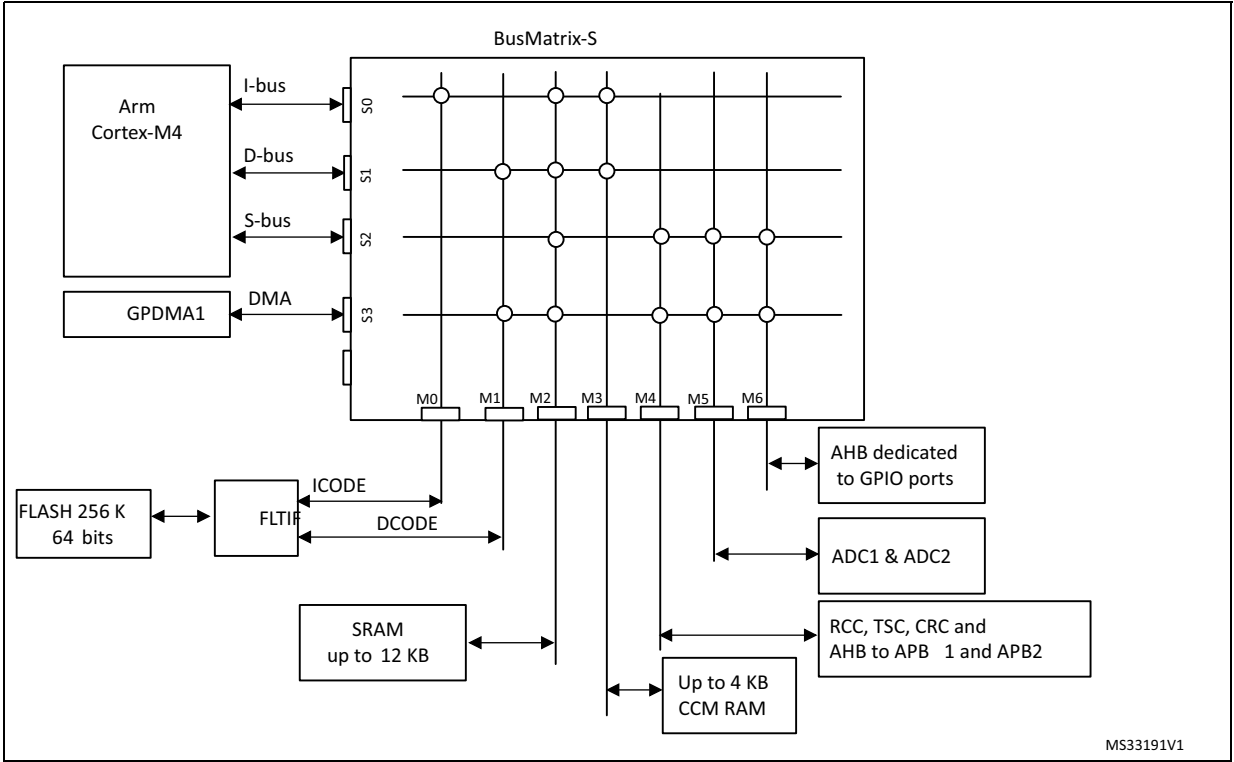
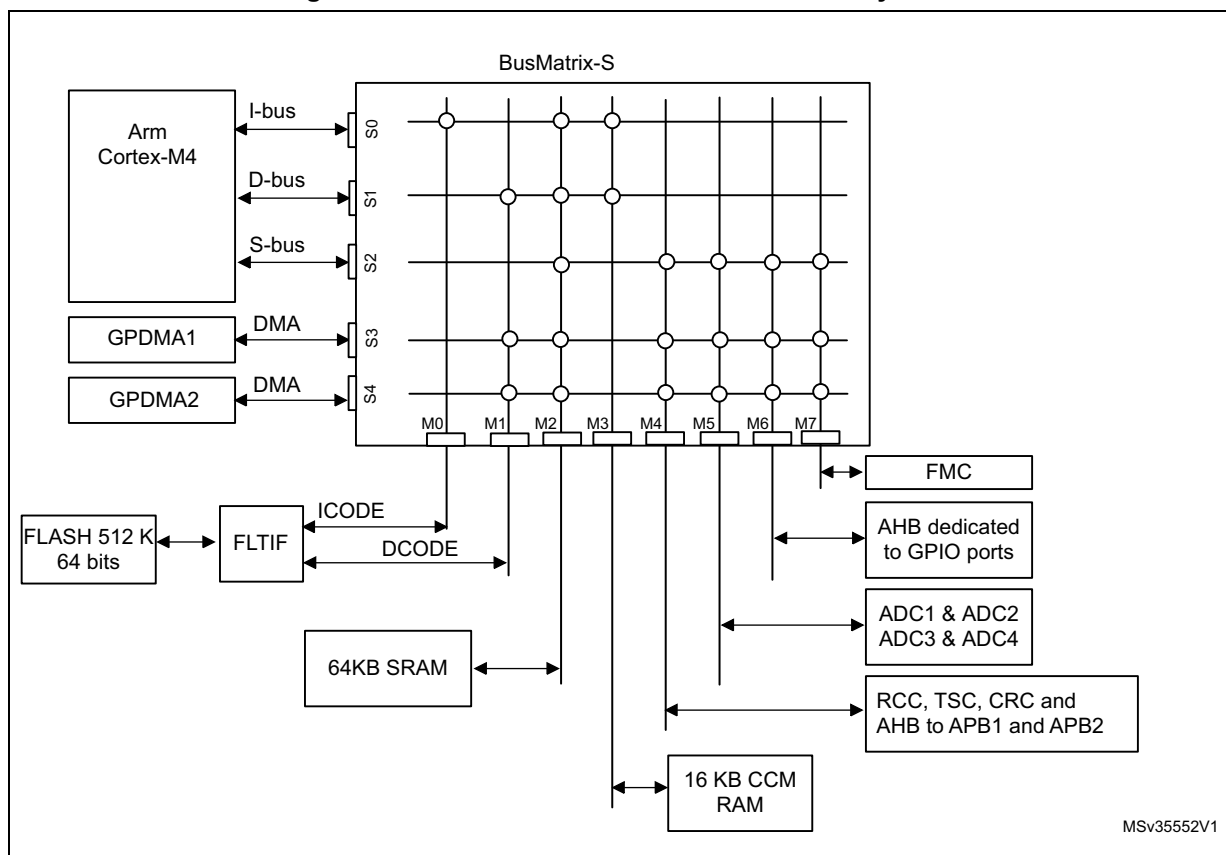


Figure 3. STM32F303xDxE and STM32F398xE system architecture



3.1.1 S0: I-bus

This bus connects the instruction bus of the Cortex[®]-M4 core to the BusMatrix. This bus is used by the core to fetch instructions. The targets of this bus are the internal Flash memory, the SRAM and the CCM SRAM.

3.1.2 S1: D-bus

This bus connects the DCode bus (literal load and debug access) of the Cortex[®]-M4 core to the BusMatrix. The targets of this bus are the internal Flash memory, the SRAM and the CCM SRAM.

3.1.3 S2: S-bus

This bus connects the system bus of the Cortex[®]-M4 core to the BusMatrix. This bus is used to access data located in the peripheral or SRAM area. The targets of this bus are the SRAM, the AHB to APB1/APB2 bridges, the AHB IO port and the ADC.

3.1.4 S3, S4: DMA-bus

This bus connects the AHB master interface of the DMA to the BusMatrix, which manages the access of different Masters to flash, SRAM, and peripherals.

3.1.5 BusMatrix

The BusMatrix manages the access arbitration between Masters. The arbitration uses a Round Robin algorithm. The BusMatrix is composed of five masters (CPU AHB, System bus, DCode bus, ICode bus, DMA1/2 bus) and seven slaves (FLITF, SRAM, AHB2GPIO and AHB2APB1/2 bridges, and ADC).

AHB/APB bridges

The two AHB/APB bridges provide full synchronous connections between the AHB and the two APB buses. APB1 is limited to 36 MHz. APB2 operates at full speed (72 MHz).

Refer to [Section 3.2: Memory organization on page 53](#) for the address mapping of the peripherals connected to this bridge.

After each device reset, all peripheral clocks are disabled (except for the SRAM and FLITF). Before using a peripheral the user has to enable its clock in the RCC_AHBENR, RCC_APB2ENR or RCC_APB1ENR register.

When a 16- or 8-bit access is performed on an APB register, the access is transformed into a 32-bit access: the bridge duplicates the 16- or 8-bit data to feed the 32-bit vector.

3.2 Memory organization

3.2.1 Introduction

Program memory, data memory, registers and I/O ports are organized within the same linear 4-Gbyte address space.

The bytes are coded in memory in Little Endian format. The lowest numbered byte in a word is considered the word's least significant byte and the highest numbered byte the most significant.

The addressable memory space is divided into eight main blocks, of 512 Mbytes each.

3.2.2 Memory map and register boundary addresses

All the memory map areas that are not allocated to on-chip memories and peripherals are considered “Reserved”. For the detailed mapping of available memory and register areas, refer to the following table.

The following table gives the boundary addresses of the peripherals available in the devices.

Table 2. STM32F303xB/C and STM32F358xC peripheral register boundary addresses⁽¹⁾

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
AHB3	0x5000 0400 - 0x5000 07FF	1 K	ADC3 - ADC4	Section 15.7 on page 416
	0x5000 0000 - 0x5000 03FF	1 K	ADC1 - ADC2	
-	0x4800 1800 - 0x4FFF FFFF	~132 M	Reserved	-
AHB2	0x4800 1400 - 0x4800 17FF	1 K	GPIOF	Section 11.4.12 on page 246
	0x4800 1000 - 0x4800 13FF	1 K	GPIOE	
	0x4800 0C00 - 0x4800 0FFF	1 K	GPIOD	
	0x4800 0800 - 0x4800 0BFF	1 K	GPIOC	
	0x4800 0400 - 0x4800 07FF	1 K	GPIOB	
	0x4800 0000 - 0x4800 03FF	1 K	GPIOA	
-	0x4002 4400 - 0x47FF FFFF	~128 M	Reserved	
AHB1	0x4002 4000 - 0x4002 43FF	1 K	TSC	Section 19.6.11 on page 510
	0x4002 3400 - 0x4002 3FFF	3 K	Reserved	-
	0x4002 3000 - 0x4002 33FF	1 K	CRC	Section 6.4.6 on page 96
	0x4002 2400 - 0x4002 2FFF	3 K	Reserved	-
	0x4002 2000 - 0x4002 23FF	1 K	Flash interface	Section 4.6 on page 85
	0x4002 1400 - 0x4002 1FFF	3 K	Reserved	-
	0x4002 1000 - 0x4002 13FF	1 K	RCC	Section 9.4.14 on page 168
	0x4002 0800 - 0x4002 0FFF	2 K	Reserved	-
	0x4002 0400 - 0x4002 07FF	1 K	DMA2	Section 13.6.7 on page 288
	0x4002 0000 - 0x4002 03FF	1 K	DMA1	
-	0x4001 8000 - 0x4001 FFFF	32 K	Reserved	-

Table 2. STM32F303xB/C and STM32F358xC peripheral register boundary addresses⁽¹⁾ (continued)

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
APB2	0x4001 4C00 - 0x4001 7FFF	13 K	Reserved	-
	0x4001 4800 - 0x4001 4BFF	1 K	TIM17	Section 23.6.18 on page 768
	0x4001 4400 - 0x4001 47FF	1 K	TIM16	
	0x4001 4000 - 0x4001 43FF	1 K	TIM15	Section 23.5.19 on page 748
	0x4001 3C00 - 0x4001 3FFF	1 K	Reserved	-
	0x4001 3800 - 0x4001 3BFF	1 K	USART1	Section 29.8.12 on page 959
	0x4001 3400 - 0x4001 37FF	1 K	TIM8	Section 20.4.27 on page 605
	0x4001 3000 - 0x4001 33FF	1 K	SPI1	Section 30.9.10 on page 1018
	0x4001 2C00 - 0x4001 2FFF	1 K	TIM1	Section 20.4.27 on page 605
	0x4001 0800 - 0x4001 2BFF	9 K	Reserved	-
	0x4001 0400 - 0x4001 07FF	1 K	EXTI	Section 14.3.13 on page 308
	0x4001 0000 - 0x4001 03FF	1 K	SYSCFG + COMP + OPAMP	Section 12.1.10 on page 262, Section 17.5.8 on page 470, Section 18.4.5 on page 492
-	0x4000 7800 - 0x4000 FFFF	34 K	Reserved	-

Table 2. STM32F303xB/C and STM32F358xC peripheral register boundary addresses⁽¹⁾ (continued)

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
APB1	0x4000 7400 - 0x4000 77FF	1 K	DAC1	Section 16.10.15 on page 444
	0x4000 7000 - 0x4000 73FF	1 K	PWR	Section 7.4.3 on page 111
	0x4000 6C00 - 0x4000 6FFF	1 K	Reserved	-
	0x4000 6800 - 0x4000 6BFF	1 K	Reserved	-
	0x4000 6400 - 0x4000 67FF	1 K	bxCAN	Section 31.9.5 on page 1058
	0x4000 6000 - 0x4000 63FF	1 K	USB SRAM 512 bytes	Section 32.6.3 on page 1093
	0x4000 5C00 - 0x4000 5FFF	1 K	USB device FS	
	0x4000 5800 - 0x4000 5BFF	1 K	I2C2	Section 28.9.12 on page 893
	0x4000 5400 - 0x4000 57FF	1 K	I2C1	
	0x4000 5000 - 0x4000 53FF	1 K	UART5	Section 29.8.12 on page 959
	0x4000 4C00 - 0x4000 4FFF	1 K	UART4	
	0x4000 4800 - 0x4000 4BFF	1 K	USART3	
	0x4000 4400 - 0x4000 47FF	1 K	USART2	
	0x4000 4000 - 0x4000 43FF	1 K	I2S3ext	Section 30.9.10 on page 1018
	0x4000 3C00 - 0x4000 3FFF	1 K	SPI3/I2S3	
	0x4000 3800 - 0x4000 3BFF	1 K	SPI2/I2S2	
	0x4000 3400 - 0x4000 37FF	1 K	I2S2ext	
	0x4000 3000 - 0x4000 33FF	1 K	IWDG	Section 25.4.6 on page 779
	0x4000 2C00 - 0x4000 2FFF	1 K	WWDG	Section 26.5.4 on page 785
	0x4000 2800 - 0x4000 2BFF	1 K	RTC	Section 27.6.20 on page 827
-	0x4000 1800 - 0x4000 27FF	4 K	Reserved	-
APB1	0x4000 1400 - 0x4000 17FF	1 K	TIM7	Section 22.4.9 on page 690
	0x4000 1000 - 0x4000 13FF	1 K	TIM6	
	0x4000 0C00 - 0x4000 0FFF	1 K	Reserved	-
	0x4000 0800 - 0x4000 0BFF	1 K	TIM4	Section 21.4.22 on page 676
	0x4000 0400 - 0x4000 07FF	1 K	TIM3	
	0x4000 0000 - 0x4000 03FF	1 K	TIM2	
-	0x2000 A000 - 3FFF FFFF	~512 M	Reserved	-
-	0x2000 0000 - 0x2000 9FFF	40 K	SRAM	-
-	0x1FFF F800 - 0x1FFF FFFF	2 K	Option bytes	-
-	0x1FFF D800 - 0x1FFF F7FF	8 K	System memory	-
-	0x1000 2000 - 0x1FFF D7FF	~256 M	Reserved	-
-	0x1000 0000 - 0x1000 1FFF	8 K	CCM SRAM	-

Table 2. STM32F303xB/C and STM32F358xC peripheral register boundary addresses⁽¹⁾ (continued)

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
-	0x0804 0000 - 0x0FFF FFFF	~128 M	Reserved	-
-	0x0800 0000 - 0x0803 FFFF	256 K	Main Flash memory	-
-	0x0004 0000 - 0x07FF FFFF	~128 M	Reserved	-
-	0x0000 000 - 0x0003 FFFF	256 K	Main Flash memory, system memory or SRAM depending on BOOT configuration	-

1. The gray color is used for reserved Flash memory addresses.

Table 3. STM32F303xD/E and STM32F398xE peripheral register boundary addresses⁽¹⁾

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
AHB4	0xA000 0400 - 0xA000 0FFF	4 K	FMC control registers	Section 10.6.9: FMC register map
	0x8000 0400 - 0x9FFF FFFF	512 M	FMC banks 3 and 4	
	0x6000 0000 - 0x7FFF FFFF	512 M	FMC banks 1 and 2	
-	0x5000 0800 - 0x5FFF FFFF	384M	Reserved	-
AHB3	0x5000 0400 - 0x5000 07FF	1 K	ADC3 - ADC4	Section 15.7 on page 416
	0x5000 0000 - 0x5000 03FF	1 K	ADC1 - ADC2	
-	0x4800 2000 - 0x4FFF FFFF	~132 M	Reserved	-
AHB2	0x4800 1C00 - 0x4800 1FFF	1 K	GPIOH	Section 11.4.12 on page 246
	0x4800 1800 - 0x4800 1BFF	1 K	GPIOG	
	0x4800 1400 - 0x4800 17FF	1 K	GPIOF	
	0x4800 1000 - 0x4800 13FF	1 K	GPIOE	
	0x4800 0C00 - 0x4800 0FFF	1 K	GPIOD	
	0x4800 0800 - 0x4800 0BFF	1 K	GPIOC	
	0x4800 0400 - 0x4800 07FF	1 K	GPIOB	
	0x4800 0000 - 0x4800 03FF	1 K	GPIOA	
-	0x4002 4400 - 0x47FF FFFF	~128 M	Reserved	

Table 3. STM32F303xD/E and STM32F398xE peripheral register boundary addresses⁽¹⁾ (continued)

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
AHB1	0x4002 4000 - 0x4002 43FF	1 K	TSC	Section 19.6.11 on page 510
	0x4002 3400 - 0x4002 3FFF	3 K	Reserved	-
	0x4002 3000 - 0x4002 33FF	1 K	CRC	Section 6.4.6 on page 96
	0x4002 2400 - 0x4002 2FFF	3 K	Reserved	-
	0x4002 2000 - 0x4002 23FF	1 K	Flash interface	Section 4.6 on page 85
	0x4002 1400 - 0x4002 1FFF	3 K	Reserved	-
	0x4002 1000 - 0x4002 13FF	1 K	RCC	Section 9.4.14 on page 168
	0x4002 0800 - 0x4002 0FFF	2 K	Reserved	-
	0x4002 0400 - 0x4002 07FF	1 K	DMA2	Section 13.6.6 on page 288
	0x4002 0000 - 0x4002 03FF	1 K	DMA1	
-	0x4001 8000 - 0x4001 FFFF	32 K	Reserved	-
APB2	0x4001 4C00 - 0x4001 4FFF	1 K	Reserved	-
	0x4001 5400 - 0x4001 7FFF	11K	Reserved	-
	0x4001 5000 - 0x4001 53FF	1 K	TIM20	Section 23.6.18 on page 768
	0x4001 4800 - 0x4001 4BFF	1 K	TIM17	
	0x4001 4400 - 0x4001 47FF	1 K	TIM16	
	0x4001 4000 - 0x4001 43FF	1 K	TIM15	Section 23.5.19 on page 748
	0x4001 3C00 - 0x4001 3FFF	1 K	SPI4	Section 30.9.10 on page 1018
	0x4001 3800 - 0x4001 3BFF	1 K	USART1	Section 29.8.12 on page 959
	0x4001 3400 - 0x4001 37FF	1 K	TIM8	Section 20.4.27 on page 605
	0x4001 3000 - 0x4001 33FF	1 K	SPI1	Section 30.9.10 on page 1018
	0x4001 2C00 - 0x4001 2FFF	1 K	TIM1	Section 20.4.27 on page 605
	0x4001 0800 - 0x4001 2BFF	9 K	Reserved	-
	0x4001 0400 - 0x4001 07FF	1 K	EXTI	Section 14.3.13 on page 308
	0x4001 0000 - 0x4001 03FF	1 K	SYSCFG + COMP + OPAMP	Section 12.1.10 on page 262 , Section 17.5.8 on page 470 , Section 18.4.5 on page 492
-	0x4000 7C00 - 0x4000 FFFF	33 K	Reserved	-

Table 3. STM32F303xD/E and STM32F398xE peripheral register boundary addresses⁽¹⁾ (continued)

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
APB1	0x4000 7800 - 0x4000 7BFF	1 K	I2C3	Section 28.9.12 on page 893
	0x4000 7400 - 0x4000 77FF	1 K	DAC1	Section 16.10.15 on page 444
	0x4000 7000 - 0x4000 73FF	1 K	PWR	Section 7.4.3 on page 111
	0x4000 6C00 - 0x4000 6FFF	1 K	Reserved	-
	0x4000 6800 - 0x4000 6BFF	1 K	Reserved	-
	0x4000 6400 - 0x4000 67FF	1 K	bxCAN	Section 31.9.5 on page 1058
	0x4000 6000 - 0x4000 63FF	1 K	USB/CAN SRAM	Section 32.6.3 on page 1093
	0x4000 5C00 - 0x4000 5FFF	1 K	USB device FS	
	0x4000 5800 - 0x4000 5BFF	1 K	I2C2	Section 28.9.1 on page 880
	0x4000 5400 - 0x4000 57FF	1 K	I2C1	
	0x4000 5000 - 0x4000 53FF	1 K	UART5	Section 29.8.12 on page 959
	0x4000 4C00 - 0x4000 4FFF	1 K	UART4	
	0x4000 4800 - 0x4000 4BFF	1 K	USART3	
	0x4000 4400 - 0x4000 47FF	1 K	USART2	
	0x4000 4000 - 0x4000 43FF	1 K	I2S3ext	Section 30.9.10 on page 1018
	0x4000 3C00 - 0x4000 3FFF	1 K	SPI3/I2S3	
	0x4000 3800 - 0x4000 3BFF	1 K	SPI2/I2S2	
	0x4000 3400 - 0x4000 37FF	1 K	I2S2ext	
	0x4000 3000 - 0x4000 33FF	1 K	IWDG	Section 25.4.6 on page 779
	0x4000 2C00 - 0x4000 2FFF	1 K	WWDG	Section 26.5.4 on page 785
	0x4000 2800 - 0x4000 2BFF	1 K	RTC	Section 27.6.20 on page 827
-	0x4000 1800 - 0x4000 27FF	4 K	Reserved	-
APB1	0x4000 1400 - 0x4000 17FF	1 K	TIM7	Section 22.4.9 on page 690
	0x4000 1000 - 0x4000 13FF	1 K	TIM6	
	0x4000 0C00 - 0x4000 0FFF	1 K	Reserved	-
	0x4000 0800 - 0x4000 0BFF	1 K	TIM4	Section 21.4.22 on page 676
	0x4000 0400 - 0x4000 07FF	1 K	TIM3	
	0x4000 0000 - 0x4000 03FF	1 K	TIM2	
-	0x2000 A000 - 3FFF FFFF	~512 M	Reserved	-
-	0x2000 0000 - 0x2000 FFFF	64 K	SRAM	-
-	0x1FFF F800 - 0x1FFF FFFF	2 K	Option bytes	-
-	0x1FFF D800 - 0x1FFF F7FF	8 K	System memory	-
-	0x1000 2000 - 0x1FFF D7FF	~256 M	Reserved	-

Table 3. STM32F303xD/E and STM32F398xE peripheral register boundary addresses⁽¹⁾ (continued)

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
-	0x1000 0000 - 0x1000 3FFF	16 K	CCM SRAM	-
-	0x0808 0000 - 0x0FFF FFFF	~128 M	Reserved	-
-	0x0800 0000 - 0x0807 FFFF	512 K	Main Flash memory	-
-	0x0008 0000 - 0x07FF FFFF	~128 M	Reserved	-
-	0x0000 000 - 0x0007 FFFF	512 K	Main Flash memory, system memory or SRAM depending on BOOT configuration	-

1. The gray color is used for reserved Flash memory addresses.

Table 4. STM32F303x6/8 and STM32F328x8 peripheral register boundary addresses⁽¹⁾

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
AHB3	0x5000 0400 - 0x5000 07FF	1 K	Reserved	-
	0x5000 0000 - 0x5000 03FF	1 K	ADC1 - ADC2	Section 15.7 on page 416
-	0x4800 1800 - 0x4FFF FFFF	~132 M	Reserved	-
AHB2	0x4800 1400 - 0x4800 17FF	1 K	GPIOF	Section 11.4.12 on page 246
	0x4800 1000 - 0x4800 13FF	1 K	Reserved	
	0x4800 0C00 - 0x4800 0FFF	1 K	GPIOD	
	0x4800 0800 - 0x4800 0BFF	1 K	GPIOC	
	0x4800 0400 - 0x4800 07FF	1 K	GPIOB	
	0x4800 0000 - 0x4800 03FF	1 K	GPIOA	
-	0x4002 4400 - 0x47FF FFFF	~128 M	Reserved	
AHB1	0x4002 4000 - 0x4002 43FF	1 K	TSC	Section 19.6.11 on page 510
	0x4002 3400 - 0x4002 3FFF	3 K	Reserved	-
	0x4002 3000 - 0x4002 33FF	1 K	CRC	Section 6.4.6 on page 96
	0x4002 2400 - 0x4002 2FFF	3 K	Reserved	-
	0x4002 2000 - 0x4002 23FF	1 K	Flash interface	Section 4.6 on page 85
	0x4002 1400 - 0x4002 1FFF	3 K	Reserved	-
	0x4002 1000 - 0x4002 13FF	1 K	RCC	Section 9.4.14 on page 168
	0x4002 0400 - 0x4002 0FFF	3 K	Reserved	-
	0x4002 0000 - 0x4002 03FF	1 K	DMA1	Section 13.6.7 on page 288
-	0x4001 8000 - 0x4001 FFFF	32 K	Reserved	-

Table 4. STM32F303x6/8 and STM32F328x8 peripheral register boundary addresses⁽¹⁾ (continued)

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
APB2	0x4001 4C00 - 0x4001 7FFF	13 K	Reserved	-
	0x4001 4800 - 0x4001 4BFF	1 K	TIM17	Section 23.6.18 on page 768
	0x4001 4400 - 0x4001 47FF	1 K	TIM16	
	0x4001 4000 - 0x4001 43FF	1 K	TIM15	Section 23.5.19 on page 748
	0x4001 3C00 - 0x4001 3FFF	1 K	Reserved	-
	0x4001 3800 - 0x4001 3BFF	1 K	USART1	Section 29.8.12 on page 959
	0x4001 3400 - 0x4001 37FF	1 K	Reserved	-
	0x4001 3000 - 0x4001 33FF	1 K	SPI1	Section 30.9.10 on page 1018
	0x4001 2C00 - 0x4001 2FFF	1 K	TIM1	Section 20.4.27 on page 605
	0x4001 0800 - 0x4001 2BFF	9 K	Reserved	-
	0x4001 0400 - 0x4001 07FF	1 K	EXTI	Section 14.3.13 on page 308
	0x4001 0000 - 0x4001 03FF	1 K	SYSCFG + COMP + OPAMP	Section 12.1.10 on page 262, Section 17.5.8 on page 470, Section 18.4.5 on page 492
-	0x4000 9C00 - 0x4000 FFFF	25 K	Reserved	-

Table 4. STM32F303x6/8 and STM32F328x8 peripheral register boundary addresses⁽¹⁾ (continued)

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
APB1	0x4000 9800 - 0x4000 9BFF	1 K	DAC2	Section 16.10.15 on page 444
	0x4000 7800 - 0x4000 97FF	8 K	Reserved	-
	0x4000 7400 - 0x4000 77FF	1 K	DAC1	Section 16.10.15 on page 444
	0x4000 7000 - 0x4000 73FF	1 K	PWR	Section 7.4.3 on page 111
	0x4000 6C00 - 0x4000 6FFF	1 K	Reserved	-
	0x4000 6800 - 0x4000 6BFF	1 K	Reserved	-
	0x4000 6400 - 0x4000 67FF	1 K	bxCAN	Section 31.9.5 on page 1058
	0x4000 5800 - 0x4000 63FF	3 K	Reserved	-
	0x4000 5400 - 0x4000 57FF	1 K	I2C1	Section 28.9.12 on page 893
	0x4000 4C00 - 0x4000 53FF	2 K	Reserved	-
	0x4000 4800 - 0x4000 4BFF	1 K	USART3	Section 29.8.12 on page 959
	0x4000 4400 - 0x4000 47FF	1 K	USART2	
	0x4000 3400 - 0x4000 43FF	4 K	Reserved	-
	0x4000 3000 - 0x4000 33FF	1 K	IWDG	Section 25.4.6 on page 779
	0x4000 2C00 - 0x4000 2FFF	1 K	WWDG	Section 26.5.4 on page 785
	0x4000 2800 - 0x4000 2BFF	1 K	RTC	Section 27.6.20 on page 827
	0x4000 1800 - 0x4000 27FF	4 K	Reserved	-
	0x4000 1400 - 0x4000 17FF	1 K	TIM7	Section 22.4.9 on page 690
	0x4000 1000 - 0x4000 13FF	1 K	TIM6	
	0x4000 0800 - 0x4000 0FFF	2K	Reserved	-
	0x4000 0400 - 0x4000 07FF	1 K	TIM3	Section 21.4.22 on page 676
	0x4000 0000 - 0x4000 03FF	1 K	TIM2	-
-	0x2000 3000 - 3FFF FFFF	~512 M	Reserved	-
-	0x2000 0000 - 0x2000 2FFF	12 K	SRAM	-
-	0x1FFF F800 - 0x1FFF FFFF	2 K	Option bytes	-
-	0x1FFF D800 - 0x1FFF F7FF	8 K	System memory	-
-	0x1000 1000 - 0x1FFF D7FF	~256 M	Reserved	-
-	0x1000 0000 - 0x1000 0FFF	4 K	CCM SRAM	-
-	0x0804 0000 - 0x0FFF FFFF	~128 M	Reserved	-
-	0x0800 0000 - 0x0800 FFFF	64 K	Main Flash memory	-

Table 4. STM32F303x6/8 and STM32F328x8 peripheral register boundary addresses⁽¹⁾ (continued)

Bus	Boundary address	Size (bytes)	Peripheral	Peripheral register map
-	0x0001 0000 - 0x07FF FFFF	~128 M	Reserved	-
-	0x0000 000 - 0x0000 FFFF	64 K	Main Flash memory, system memory or SRAM depending on BOOT configuration	-

1. The gray color is used for reserved Flash memory addresses.

3.3 Embedded SRAM

STM32F303xB/C and STM32F358xC devices feature up to 48 Kbytes of static SRAM. It can be accessed as bytes, halfwords (16 bits) or full words (32 bits):

- Up to 40 Kbytes of SRAM that can be addressed at maximum system clock frequency without wait states and can be accessed by both CPU and DMA;
- 8 Kbytes of CCM SRAM. It is used to execute critical routines or to access data. It can be accessed by the CPU only. No DMA accesses are allowed. This memory can be addressed at maximum system clock frequency without wait state.

STM32F303xD/E and STM32F398xE devices feature up to 80 Kbytes of static SRAM. It can be accessed as bytes, halfwords (16 bits) or full words (32 bits):

- Up to 64 Kbytes of SRAM that can be addressed at maximum system clock frequency without wait states and can be accessed by both CPU and DMA;
- 16 Kbytes of CCM SRAM. It is used to execute critical routines or to access data. It can be accessed by the CPU only. No DMA accesses are allowed. This memory can be addressed at maximum system clock frequency without wait state.

STM32F303x6/8 and STM32F328x8 devices feature the same memory but only up to 16 Kbytes of static SRAM: up to 12 Kbytes of SRAM and 4 Kbytes of CCM SRAM.

3.3.1 Parity check

On the STM32F303xB/C and STM32F358xC devices, for the 40-Kbyte SRAM, a parity check is implemented only on the first 16 Kbytes.

The SRAM parity check is disabled by default. It is enabled by the user, when needed, using an option bit.

On the STM32F303x6/8 and STM32F328x8 devices, the parity check is implemented on all of the SRAM and CCM SRAM.

On the STM32F303xD/E and STM32F398xE devices, the parity check is implemented on the first 32 Kbytes of SRAM and on the whole CCM SRAM

The data bus width of the SRAM supporting the parity check is 36 bits because 4 bits are available for parity check (1 bit per byte) in order to increase memory robustness, as required for instance by Class B or SIL norms.

The parity bits are computed on data and address and stored when writing into the SRAM. Then, they are automatically checked when reading. If one bit fails, an NMI is generated if the SRAM parity check is enabled. The same error can also be linked to the Break input of

TIMER 20, 1, 8, 15, 16 and 17, by setting the SRAM_PARITY_LOCK control bit in the SYSCFG configuration register 2 (SYSCFG_CFGR2). In case of parity error, the SRAM Parity Error flag (SRAM_PEF) is set in the SYSCFG configuration register 2 (SYSCFG_CFGR2). For more details, please refer to the SYSCFG configuration register 2 (SYSCFG_CFGR2).

The BYP_ADD_PAR bit in SYSCFG_CFGR2 register can be used to prevent an unwanted parity error to occur when the user programs a code in the RAM at address 0x2XXXXXXX (address in the address range 0x20000000-0x20002000) and then executes the code from RAM at boot (RAM is remapped at address 0x00).

3.3.2 CCM SRAM write protection

The CCM SRAM is write protected with a page granularity of 1 Kbyte.

Table 5. CCM SRAM organization

Page number	Start address	End address
Page 0	0x1000 0000	0x1000 03FF
Page 1	0x1000 0400	0x1000 07FF
Page 2	0x1000 0800	0x1000 0BFF
Page 3	0x1000 0C00	0x1000 0FFF
Page 4 ⁽¹⁾	0x1000 1000	0x1000 13FF
Page 5 ⁽¹⁾	0x1000 1400	0x1000 17FF
Page 6 ⁽¹⁾	0x1000 1800	0x1000 1BFF
Page 7 ⁽¹⁾	0x1000 1C00	0x1000 1FFF
Page 8 ⁽²⁾	0x1000 2000	0x1000 23FF
Page 9 ⁽²⁾	0x1000 2400	0x1000 27FF
Page 10 ⁽²⁾	0x1000 2800	0x1000 2BFF
Page 11 ⁽²⁾	0x1000 2C00	0x1000 2FFF
Page 12 ⁽²⁾	0x1000 3000	0x1000 33FF
Page 13 ⁽²⁾	0x1000 3400	0x1000 37FF
Page 14 ⁽²⁾	0x1000 3800	0x1000 3BFF
Page 15 ⁽²⁾	0x1000 3C00	0x1000 3FFF

1. Only on STM32F303xB/C/D/E and STM32F358xC devices.

2. Only on STM32F303xD/E and STM32F398xE devices.

The write protection can be enabled in the CCM SRAM protection register (SYSCFG_RCR) in the SYSCFG block. This is a register with write '1' once mechanism, which means by writing '1' on a bit it will setup the write protection for that page of SRAM and it can be removed/cleared by a system reset only. For more details please refer to the SYSCFG section.

3.4 Flash memory overview

The Flash memory is composed of two distinct physical areas:

- The main Flash memory block. It contains the application program and user data if necessary.
- The information block. It is composed of two parts:
 - Option bytes for hardware and memory protection user configuration.
 - System memory which contains the proprietary boot loader code. Please, refer to [Section 4: Embedded flash memory](#) for more details.

Flash memory instructions and data access are performed through the AHB bus. The prefetch block is used for instruction fetches through the ICode bus. Arbitration is performed in the Flash memory interface, and priority is given to data access on the DCode bus. It also implements the logic necessary to carry out the Flash memory operations (Program/Erase) controlled through the Flash registers.

3.5 Boot configuration

In the STM32F3xx, three different boot modes can be selected through the BOOT0 pin and nBOOT1 bit in the User option byte, as shown in the following table:

Table 6. Boot modes

Boot mode selection		Boot mode	Aliasing
nBOOT1	BOOT0		
x	0	Main Flash memory	Main flash memory is selected as boot area
1	1	System memory	System memory is selected as boot area
0	1	Embedded SRAM	Embedded SRAM (on the DCode bus) is selected as boot area

The values on both BOOT0 pin and nBOOT1 bit are latched on the 4th rising edge of SYSCLK after a reset.

It is up to the user to set the nBOOT1 and BOOT0 to select the required boot mode. The BOOT0 pin and nBOOT1 bit are also resampled when exiting from Standby mode. Consequently they must be kept in the required Boot mode configuration in Standby mode. After this startup delay has elapsed, the CPU fetches the top-of-stack value from address 0x0000 0000, then starts code execution from the boot memory at 0x0000 0004. Depending on the selected boot mode, main Flash memory, system memory or SRAM is accessible as follows:

- Boot from main Flash memory: the main Flash memory is aliased in the boot memory space (0x0000 0000), but still accessible from its original memory space (0x0800 0000). In other words, the Flash memory contents can be accessed starting from address 0x0000 0000 or 0x0800 0000.
- Boot from system memory: the system memory is aliased in the boot memory space (0x0000 0000), but still accessible from its original memory space (0x1FFF D800).
- Boot from the embedded SRAM: the SRAM is aliased in the boot memory space (0x0000 0000), but it is still accessible from its original memory space (0x2000 0000).

3.5.1 Embedded boot loader

The embedded boot loader is located in the System memory, programmed by ST during production. It is used to reprogram the Flash memory through:

- USART1 (PA9/PA10), USART2 (PD5/PD6) or USB (DFU) on STM32F303xB/C devices,
- USART1 (PA9/PA10), USART2 (PD5/PD6), I2C1 (PB6/PB7) on STM32F358xC devices,
- USART1 (PA9/PA10), USART2 (PA2/PA3), I2C1 (PB6/PB7) on STM32F303x6/8 and STM32F328x8 devices,
- USART1 (PA9/PA10), USART2 (PA2/PA3) or USB (DFU) on STM32F303xD/E devices.
- USART1 (PA9/PA10) or USART2 (PA2/PA3) or I2C1 (PB6/PB7) or I2C3 (PA8/PB5) on STM32F398xE.

Note: For more details see the corresponding datasheets.